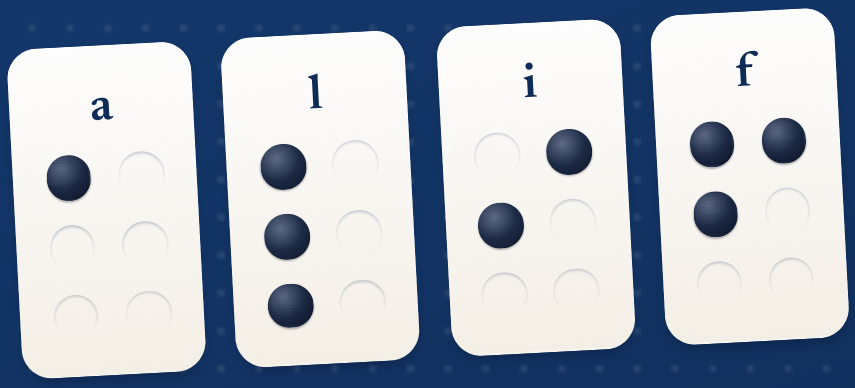
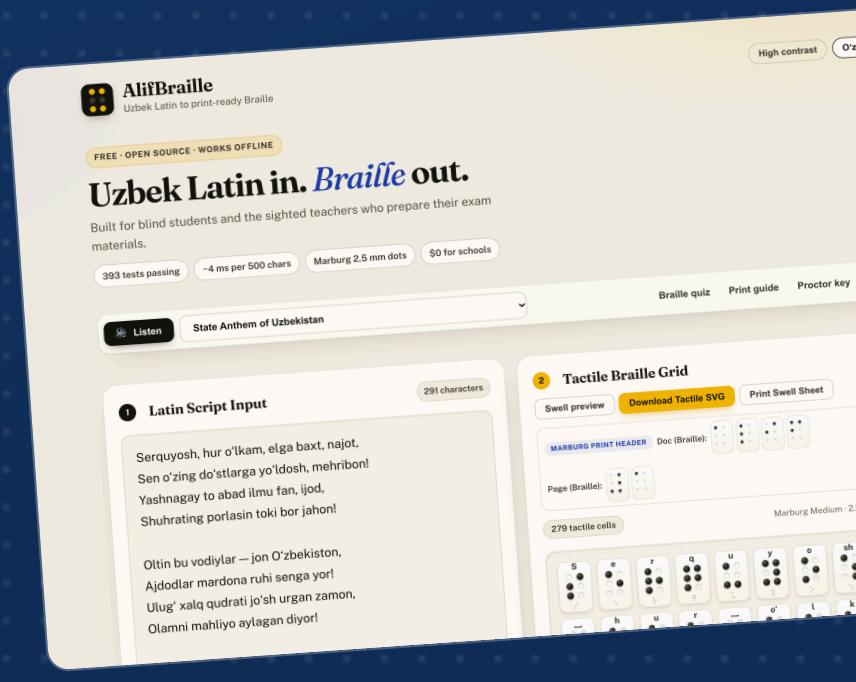
STUDENT IMPACT STATEMENT · SXSW EDU 2027

Same exam. *Same alphabet.*

How I built AlifBraille, a free tool that turns Uzbek Latin into correct, print-ready Braille, so blind students can prepare in the alphabet their exam is written in.



"alif" in six-dot Braille. Alif is the first letter, and this is a first step.



Elshod Kenjayev
Grade 11, Presidential School in Tashkent
Tashkent, Uzbekistan · October 2026

braille.elshod.xyz
braille.elshod.xyz/pitch
github.com/Elshod09K/alifbraille

01 · WHY I STARTED

A small change for me. A big one for them.

Everything in my classes at the Presidential School in Tashkent is in Latin now. Uzbekistan has been moving its schools, textbooks and university entrance exams from Cyrillic to the Uzbek Latin alphabet, and for me the switch was easy: new books, a few spelling habits to unlearn.

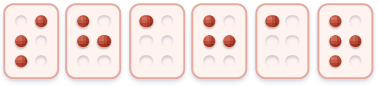
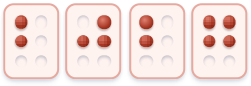
Then I asked a question I couldn't let go of. What about students who read with their fingers? Most tactile books in school libraries were produced in Cyrillic Braille. I couldn't find a free tool that turns Uzbek Latin into correct Braille. And many subject teachers (history, literature, maths) don't read Braille at all, so they have no way to check a worksheet before they hand it out. They wait for a specialist transcriber, or they guess.

So a blind student preparing for the same exam as me gets fewer practice papers, later, and often in the wrong script. I thought that was something I could actually fix, so I started building.

02 · WHAT GOES WRONG TODAY

One wrong cell can change the answer.

Uzbek Latin looks simple until you try to convert it. O‘ and G‘ are letters, but the mark after them looks like an apostrophe, and people type it in at least five different ways. Sh and ch are single letters written with two characters. Numbers need a number sign in front of them, or the reader sees letters. Convert letter by letter and you get this:

WORD	LETTER BY LETTER	ALIFBRAILLE	WHAT GOES WRONG
o‘g‘il son			The apostrophes become punctuation. In Uzbek, o’ and g’ are letters.
shahar city			sh is split into two cells, but it is one Uzbek letter.
2027 a year			Without the number sign, a reader sees the letters b, j, b, g.

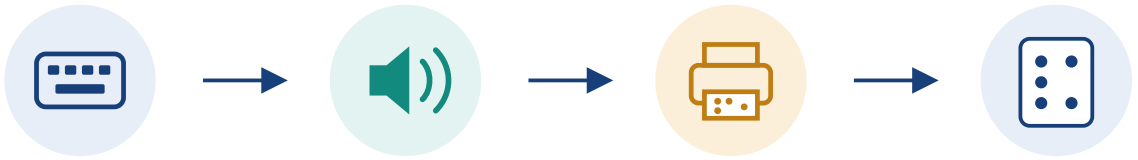
The Braille in the AlifBraille column is real output from the engine. Red cells show the mistakes a letter-by-letter converter makes.

On an exam paper, a wrong cell is a wrong answer that has nothing to do with what the student knows.

03 · WHAT I BUILT

AlifBraille, in four steps.

AlifBraille is a web app. You open it in a browser, type or paste Uzbek Latin, and the Braille appears as you type. There's no account and no server, and once it has loaded it keeps working without internet, which matters on old school computers. I wrote it in TypeScript and released it under the MIT license, so any school can use it for free.

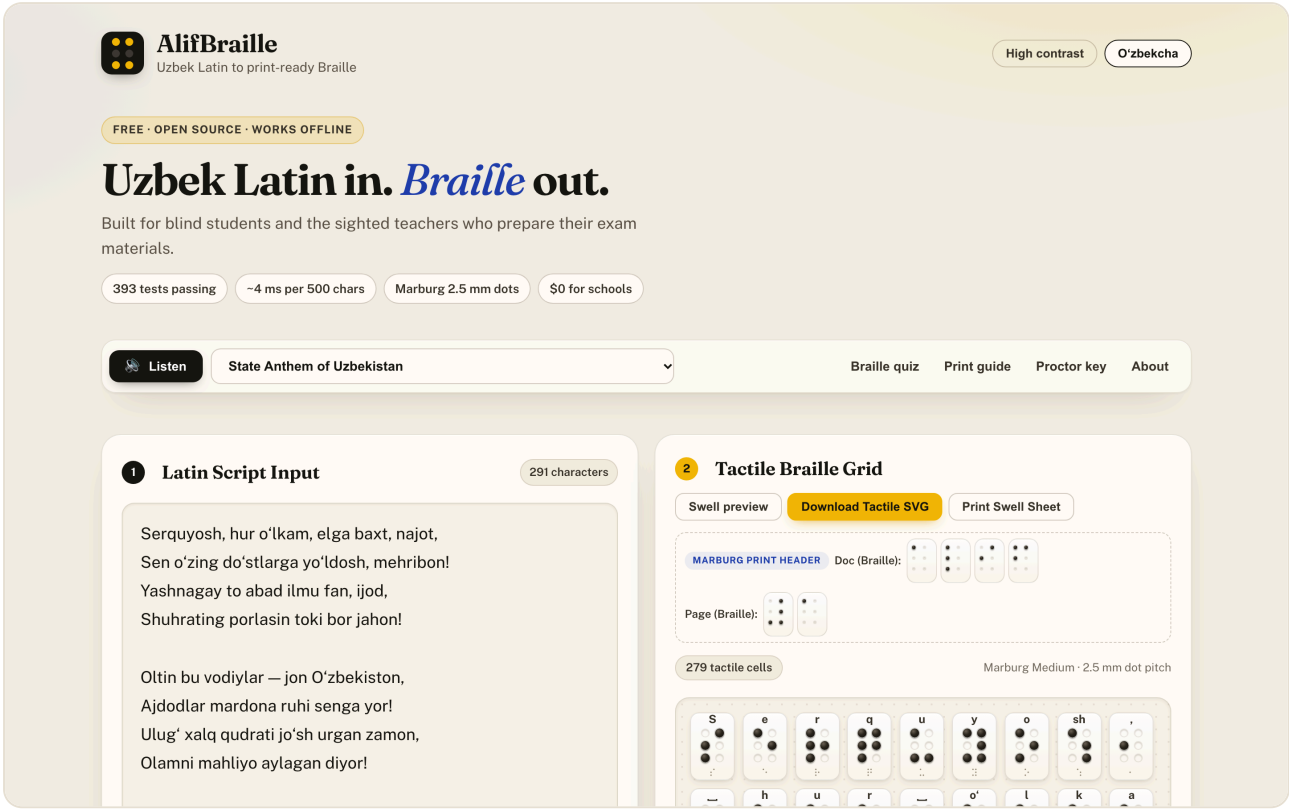


1. Type
Uzbek Latin text

2. Listen & check
each cell lights up

3. Print & heat
swell paper, 1:1 scale

4. Read by touch
same content as classmates



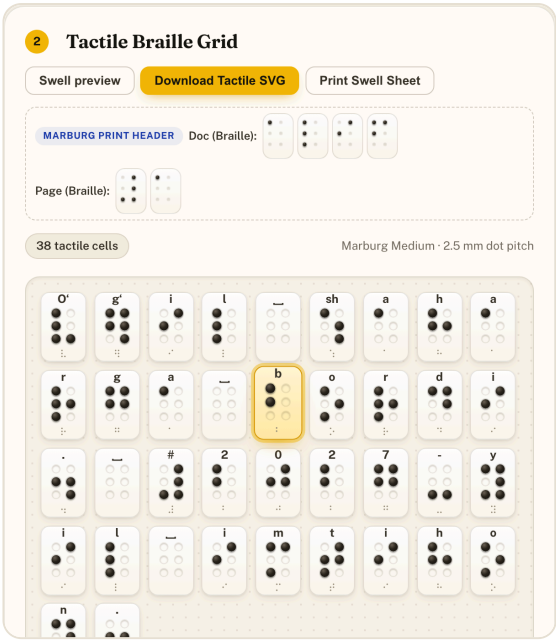
04 · BUILT SO A TEACHER CAN CHECK IT

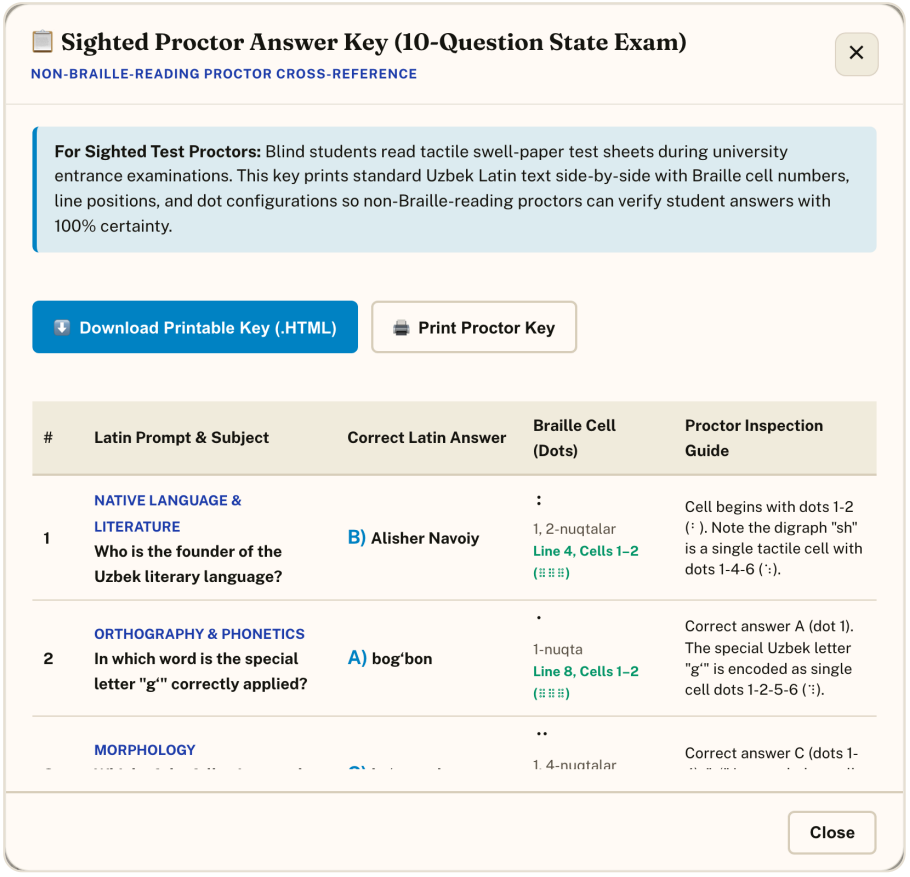
You don't need to read Braille to trust it.

This is the part I care about most. If a teacher can't read Braille, a Braille tool is only as useful as their trust in it. So AlifBraille reads the text aloud and lights up each cell as it says it. The teacher hears a letter and sees exactly which cell belongs to it.

For tests I added two more things: a ten-question template in the format of the state entrance exam, and an answer key for proctors. The key puts each question next to its Braille and gives the exact line and cell where the answer starts. In question 1, for example, the correct answer, B) *Alisher Navoiy*, starts on line 4, cells 1 to 2. A proctor who has never learned Braille can still mark the paper.

There's also a small quiz that goes the other way, from Braille back to letters, for teachers who want to start learning themselves.





The proctor answer key for the ten-question exam template.

05 · GETTING THE PAPER RIGHT

Accurate to the millimetre, before anything is heated.

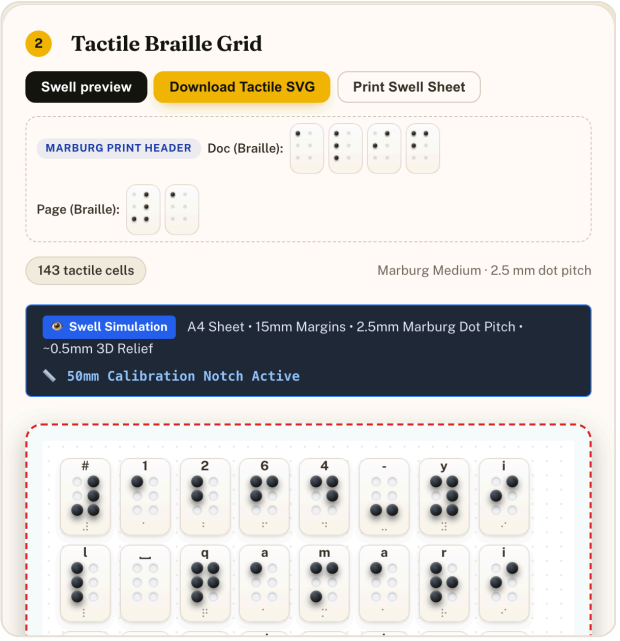
Raised-dot sheets are usually made on swell paper. You print black ink on special coated paper, run it through a heater, and the paper puffs up under the ink. It's much cheaper than a Braille embosser. But the paper is imported and expensive, and the print has to be exactly to scale.

2.5 mm	0.75 mm	10 mm	15 mm
between dots	dot radius	between lines	page margins

AlifBraille exports A4 sheets with Marburg Medium spacing. The catch is that many printers quietly shrink pages to fit, often to 94%. At 94%, 2.5 mm becomes about 2.35 mm, and once the sheet has been heated there's no fixing it.

> **So every sheet has a 50 mm line printed on it.** Hold an ordinary ruler against it before heating. If it isn't 50 mm, don't heat the sheet.

There's also a preview mode that shows the margins and raised dots on screen first, for teachers who don't have the heater in front of them.



06 · WHAT I CAN PROVE, AND WHAT I CAN'T YET

Honest about where it stands.

Everything in the blue band below can be checked in the code and its test suite.

393	3.6 ms	<0.5 s	18:1
automated tests passing, across 14 suites	to update the Braille for a 500-character passage	for the app to load	contrast in high-contrast mode

The tests cover the spelling rules, the Braille conversion, the exact dot geometry of the printed sheets, the read-aloud sync, the quiz and keyboard accessibility. I also built six sample texts to test real school material:

- State Anthem of Uzbekistan
- History exam: Amir Temur
- Alisher Navoi's sayings
- Beginner Braille drills
- Constitution: right to education
- 10-question state exam

NOT DONE YET

I haven't tested AlifBraille with blind students or their teachers. I don't have reading speeds, error rates or time saved, because I haven't measured them, and I'm not going to guess. That's what the pilot is for.

07 · WHAT HAPPENS NEXT

From a working tool to a real classroom.

NOW · DONE

A working, tested, open-source app

Exact Marburg geometry, the exam template, the proctor key and the reverse quiz. Live at braille.elshod.xyz.

NEXT

A pilot with a school for visually impaired students in Tashkent

Teachers make worksheets from their own lessons. Students read the swell-paper sheets and tell me what's hard to read or wrong. For the first time, I'll measure how readable the sheets are and how long a teacher needs to prepare one.

THEN

Fix, publish, expand

Fix what the pilot finds, publish free practice packets, and offer AlifBraille to specialized schools in other regions of Uzbekistan and in Karakalpakstan.

LATER

Review and new alphabets

Ask Braille specialists and education authorities to review the letter mapping and the print workflow, build exam-style practice packets with matching proctor keys, and extend the rules to related Latin alphabets like Karakalpak and Kazakh.

08 · WHY SXSW EDU

Why I'm bringing this to Austin.

When a country changes its alphabet, people who read Braille are easy to forget. I want to put that problem in front of people who work on accessibility and special education, and hear what they think is wrong with my approach before the pilot starts.

I'd also like other students from Central Asia to see that a student can build something useful for disability inclusion with careful engineering and no funding. And I hope to meet people who work on Braille for languages the big tools don't cover yet.

No student should be locked out of learning because of the alphabet their books are printed in.

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Try it live